

Midterm Exam

Course Title	Linear Algebra		Course Code	ELE0205	
Semester	First Semester		Academic Year	2025-2026	
Exam Date	December , 1 st ,2025	Day	Monday	Time	1:30-2:20

اسم الطالب	ANSWERS	الرقم الجامعي	الشعبة	اسم المدرس
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GENERAL INSTRUCTIONS

1. Students are not allowed to leave the exam room during the first 30 minutes of the exam.
 2. Late students are not permitted to complete their exams after the first 30 minutes unless authorized by the proctor (no students left the exam session), knowing that no additional time will be granted.
 3. Students must use a blue or black pen. Pencils are not allowed.
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 5. All electronic devices (e.g., cell phones) are prohibited.
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 9. Write the full information required on this page before commencing your exam.
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 11. Reported student cheating cases, misbehavior, or other student misconduct on examination sessions will be reported to the Investigations Committee as per internal bylaws.
- I have understood and abide with the exam policy. Student's signature: _____

Question No.	Question Type	CILOs assessed	Marks Weight	Scored Marks	Marker signature	Student Score
Part 1: 1-2,4,7	Multiple Choice	CILO 1	8 (26.67%)			
Part 1: 3,8-9	Multiple Choice	CILO 4	6 (20%)			
Part 1: 10	Multiple Choice	CILO 8	2 (6.67%)			
Part 1: 5,6	Multiple Choice	CILO 3	4 (13.33%)			
Part 2: 1	Essay Questions	CILO 9	4 (13.33%)			
Part 2: 2	Essay Questions	CILO 8	6 (20%)			
Total			30 (100%)			

Moderation Review

Comments	
Marks Reviewed by	Signature & Date

جدول الاجابات

Question	1	2	3	4	5	6	7	8	9	10
Answer	✓	B	A	C	D	D	C	B	C	B

Part 1 : Multiple Choice Questions

1. If $A = [a_{ij}]$ is 2×2 matrix such that $a_{ij} = \begin{cases} i^2 & , i < j \\ i^2 + j^2 & , i = j \\ 2 & , i > j \end{cases}$

then $A =$

A. $\begin{pmatrix} 2 & 1 \\ 2 & 8 \end{pmatrix}$

B. $\begin{pmatrix} 8 & 1 \\ 2 & 2 \end{pmatrix}$

C. $\begin{pmatrix} 2 & 1 \\ 8 & 2 \end{pmatrix}$

D. $\begin{pmatrix} 2 & 2 \\ 1 & 8 \end{pmatrix}$

2. Given that $C = \begin{pmatrix} 1 & 2 & -1 \\ 3 & 4 & 1 \\ 5 & 2 & 1 \end{pmatrix} \begin{pmatrix} 1 & 3 \\ -1 & 1 \\ 2 & 4 \end{pmatrix}$ then $c_{22} =$

A. 21

B. 17

C. 1

D. 5

3. $\begin{vmatrix} 2 & 1 & -1 \\ 0 & 0 & 4 \\ 0 & 2 & 1 \end{vmatrix} =$

A. -16

B. 4

C. -4

D. 16

4. If $A = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$ then $A^6 =$

A. $\begin{pmatrix} 6 & 6 \\ 6 & 6 \end{pmatrix}$

B. $\begin{pmatrix} 16 & 16 \\ 16 & 16 \end{pmatrix}$

C. $\begin{pmatrix} 32 & 32 \\ 32 & 32 \end{pmatrix}$

D. $\begin{pmatrix} 64 & 64 \\ 64 & 64 \end{pmatrix}$

5. The system of the linear equations $\begin{cases} x + y = 5 \\ k^2 y - y = k + 1 \end{cases}$ has no solution when $k =$

A. -1

B. 2

C. -2

D. 1

6. Given that $AX = B$ is a system of linear equations such that $\begin{pmatrix} 1 & -1 & 1 \\ 0 & 1 & 2 \\ 2 & -1 & 1 \end{pmatrix} A = I$ and

$$B = \begin{pmatrix} 2 \\ 3 \\ 1 \end{pmatrix}$$

then $X =$

A. $\begin{pmatrix} 0 \\ 2 \\ 5 \end{pmatrix}$

B. $\begin{pmatrix} 4 \\ 5 \\ 2 \end{pmatrix}$

C. $\begin{pmatrix} 2 \\ 4 \\ 5 \end{pmatrix}$

D. $\begin{pmatrix} 0 \\ 5 \\ 2 \end{pmatrix}$

7. If A and B are two square matrices of the same size then $(A - B)(A + B) =$

A. $A^2 - B^2$

B. $A^2 - AB + BA - B^2$

C. $A^2 + AB - BA - B^2$

D. $B^2 - A^2$

8. If A and B are two 4×4 matrices such that $\det(A) = 3$ and $\det(B) = 5$

then $\det(2AB^T) =$

A. 30

B. 240

C. -30

D. -240

9. If A is a square matrix such that $A^3 = 0$ then $(I + A)^{-1} =$

A. $A + A^2$

B. $I - A$

C. $I - A + A^2$

D. $I + A + A^2$

10. If $A \operatorname{adj}(A) = \begin{pmatrix} 4 & x-3 \\ 0 & 2y \end{pmatrix}$ then $\det(A) + 2x + y =$

A. 10

B. 12

C. 15

D. 17

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Question	1	2	3	4	5	6	7	8	9	10
Answer	✓	A	B	D	A	C	B	D	D	C

Part 1 : Multiple Choice Questions

(20 Points)

1) If $A = [a_{ij}]$ is 2×2 matrix such that $a_{ij} = \begin{cases} 2 & , i < j \\ i^2 + j^2 & , i = j \\ i^2 & , i > j \end{cases}$

then $A =$

A) $\begin{pmatrix} 2 & 4 \\ 1 & 8 \end{pmatrix}$

☒ B) $\begin{pmatrix} 1 & 2 \\ 4 & 8 \end{pmatrix}$

C) $\begin{pmatrix} 8 & 1 \\ 4 & 2 \end{pmatrix}$

D) $\begin{pmatrix} 2 & 4 \\ 8 & 2 \end{pmatrix}$

2) Given that $C = \begin{pmatrix} 1 & 2 & -1 \\ 3 & 4 & 1 \\ 5 & 2 & 1 \end{pmatrix} \begin{pmatrix} 1 & 3 \\ -1 & 1 \\ 2 & 4 \end{pmatrix}$ then $c_{32} =$

☒ A) 21

B) 17

C) 1

D) 5

3) $\begin{vmatrix} 2 & 1 & -1 \\ 0 & 2 & 1 \\ 0 & 0 & 4 \end{vmatrix} =$

A) -4

☒ B) 16

C) -16

D) 4

4) If $A = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$ then $A^7 =$

A) $\begin{pmatrix} 6 & 6 \\ 6 & 6 \end{pmatrix}$

☒ B) $\begin{pmatrix} 16 & 16 \\ 16 & 16 \end{pmatrix}$

C) $\begin{pmatrix} 32 & 32 \\ 32 & 32 \end{pmatrix}$

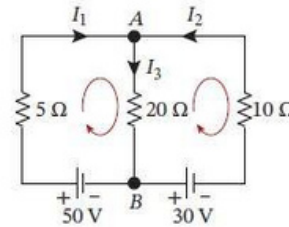
D) $\begin{pmatrix} 64 & 64 \\ 64 & 64 \end{pmatrix}$

- 5) The system of the linear equations $\begin{cases} x+y=5 \\ k^2 y-y=k+1 \end{cases}$ has infinite solution when $k =$
☒ A) -1 B) 2 C) -2 D) 1
- 6) Given that $AX=B$ is a system of linear equations such that $\begin{pmatrix} 1 & -1 & 1 \\ 0 & 1 & 2 \\ 2 & -1 & 1 \end{pmatrix} A=I$ and $B = \begin{pmatrix} 3 \\ 2 \\ 1 \end{pmatrix}$ then $X =$
 A) $\begin{pmatrix} 0 \\ 2 \\ 5 \end{pmatrix}$ B) $\begin{pmatrix} 4 \\ 5 \\ 2 \end{pmatrix}$ ☒ C) $\begin{pmatrix} 2 \\ 4 \\ 5 \end{pmatrix}$ D) $\begin{pmatrix} 0 \\ 5 \\ 2 \end{pmatrix}$
- 7) If A and B are two square matrices of the same size then $(A+B)(A-B) =$
 A) $A^2 - B^2$ B) $A^2 + AB - BA - B^2$ C) $A^2 + AB - BA - B^2$ D) $B^2 - A^2$
☒ A) $A^2 - AB + BA - B^2$
- 8) If A and B are two 4×4 matrices such that $\det(A) = 3$ and $\det(B) = -5$ then $\det(2AB^T) =$
 A) 30 B) -30 C) 240 ☒ D) -240
- 9) If A is a square matrix such that $A^3 = 0$ then $(I-A)^{-1} =$
 A) $A+A^2$ B) $I-A$ C) $I-A+A^2$ ☒ D) $I+A+A^2$
- 10) If $A \operatorname{adj}(A) = \begin{pmatrix} 4 & x-3 \\ 0 & 2y-2 \end{pmatrix}$ then $\det(A) + 3x + y =$
 A) 10 B) 12 ☒ C) 15 D) 17
 $4 + 9 + 2 = 15$

1. Applying Kirchoff Laws on the electrical circuit in the figure, give the given system of linear equations

This picture, from the textbook

$$\begin{aligned} I_1 + I_2 - I_3 &= 0 \\ +4I_3 &= 10 \\ I_2 + 2I_3 &= -3 \end{aligned}$$



Using Cramer rule,

find I_3 only.

(4Points)

$$I_3 = \frac{\begin{vmatrix} 1 & 1 & 0 \\ 1 & 0 & 10 \\ 0 & 1 & -1 \end{vmatrix}}{\begin{vmatrix} 1 & 1 & 0 \\ 1 & 0 & 10 \\ 0 & 1 & -1 \end{vmatrix}} = \frac{-7}{-7}$$

$$= \frac{-7}{-7}$$

$$\therefore I_3 = 1A$$

1 point

2 points

1 point

2. Given that $\begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} = 7$, then

(6 Points)

a. $\begin{vmatrix} a & b & c \\ 2d & 2e & 2f \\ g & h & i \end{vmatrix} = \dots\dots\dots 14 \dots\dots\dots$

2 points

b. $\begin{vmatrix} a & c & b \\ d & f & e \\ g & i & h \end{vmatrix} = \dots\dots\dots -7 \dots\dots\dots$

2 points

c. $\begin{vmatrix} a & b & c \\ 5a & 5b & 5c \\ d & e & f \end{vmatrix} = \dots\dots\dots 7 \dots\dots\dots$

2 points